

MEEG432 Project 2: XFOIL - 2D airfoil performance

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TASK 1: NACA 2018

Below in Figure 1 is a plot of the pressure distribution for two different angles of attack (AoA), 0 degrees and 10 degrees for a NACA 2018 foil. For 0 degrees angle of attack, there is positive pressure at the leading edge, with a maximum negative pressure occurring around 0.1 chord lengths into the airfoil, gradually reaching 0 pressure at the trailing edge. For 10 degrees angle of attack, there is both positive pressure and really high negative pressure at the leading edge, where the negative pressure is most likely occurring at the bottom face of the foil. Then, from both the top and bottom, the pressure gradually decreases to 0.

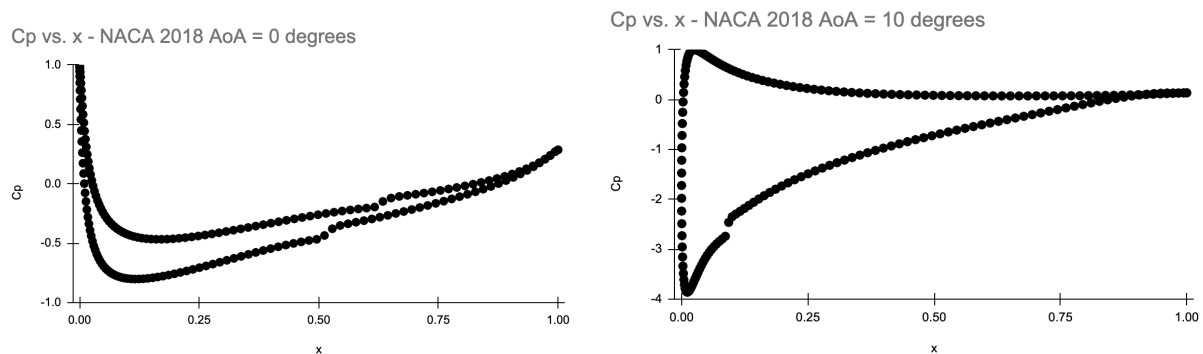


Figure 1: Plot of C_p vs x for NACA 2018 foil with angle of attack of 0 degrees (left) and 10 degrees (right)

Table 1 below shows values for coefficients of lift, drag, and moments for both angles of attack.

Table 1. Tabulated coefficients for angle of attack of 0 and 10 degrees.

α	C_l	C_d	C_m
0°	0.1884	0.00722	-0.0128
10°	1.1955	0.0028	0.01573

Below in Figure 2 are screenshots from X-Foil showing the boundary layer separation for both angles of attack. For 0 degrees angle of attack, there is very little flow separation, which occurs right at the trailing edge. For 10 degrees angle of attack, the flow starts to separate around 0.75 chord lengths in, with very large separation occurring at the trailing edge. Flow separation creates drag which will decrease performance. This means that 10 degrees angles of attack performs worse than 0 degrees angle of attack.

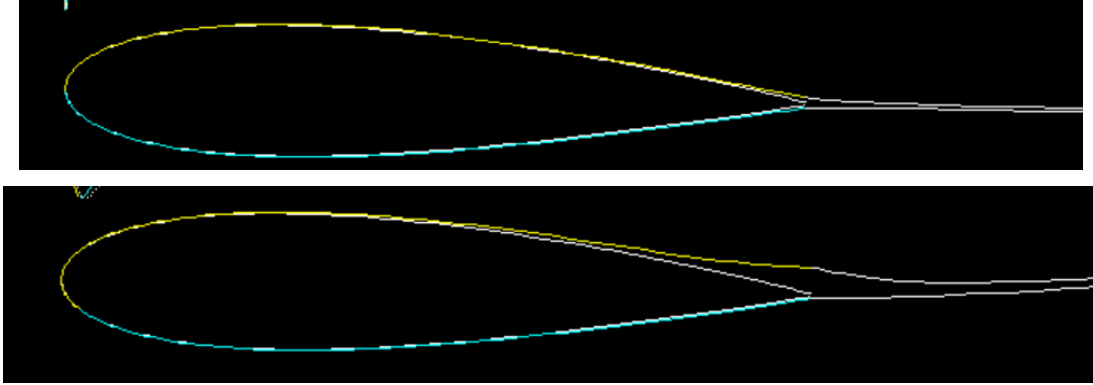


Figure 2: X-Foil screenshots of boundary layer development for the NACA 2018 airfoil at angle of attack of 0 degrees (top) and angle of attack of 10 degrees (bottom)

Figure 3 below shows plots of α vs 4 different coefficients: lift, drag, moment, and lift to drag ratio. Lift after separation is slightly reduced. The peak lift to drag ratio is 81.4394 which occurs at 8 degrees angle of attack. The slope of the lift curve should be close to 2π radians due to thin airfoil theory. The accuracy is reduced due to viscous effects.

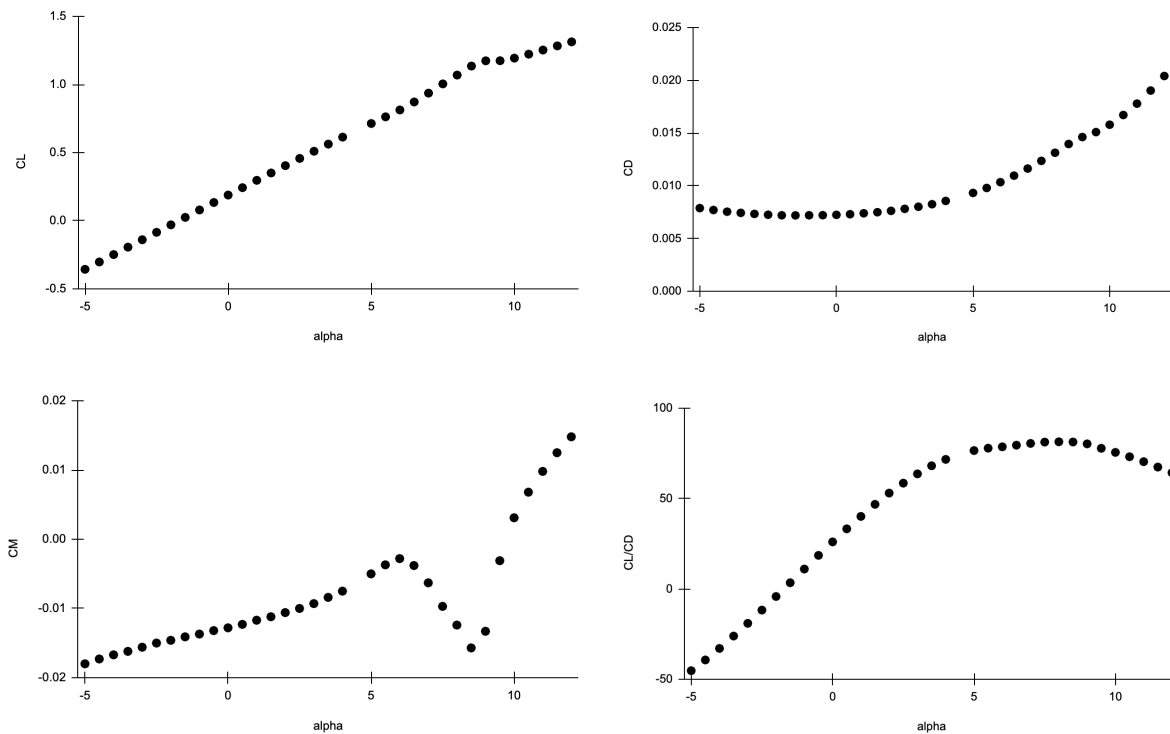


Figure 4: Performance of NACA 2018 airfoil as plots of α vs lift coefficient (top left), α vs drag coefficient (top right), α vs moment coefficient (bottom left), and α vs lift-to-drag ratio (bottom right)

TASK 2: MAXIMIZING NACA PERFORMANCE

Figure 4 below shows the performance of different NACA24XX foils. As the airfoil gets thicker, the peak performance occurs at a higher angle of attack. This is because thicker airfoils have less flow separation at higher angles of attack, which reduces drag, therefore, increasing performance. The highest performance happens with a NACA2414 at 5.5 degrees angles of attack, resulting in a C_l/C_d of 105.41.

CL/CD vs Alpha

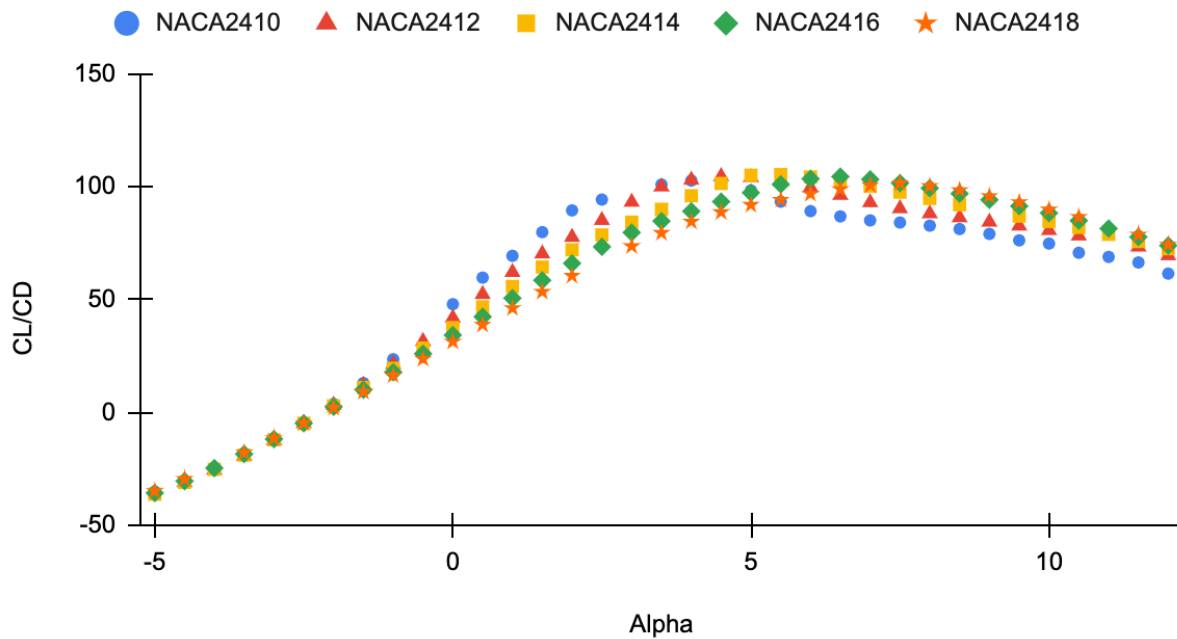


Figure 5: Performance of NACA24XX airfoils

TASK 3: CUSTOM FOIL SHAPE

Below in Figure 5 is a MATLAB plot of a custom foil and a NACA4412 foil. My foil has less camber, with the max camber occurring at 0.2 chord lengths, while the NACA4412's max camber occurs at 0.4 chord lengths. My custom foil is also much thicker than the NACA4412.

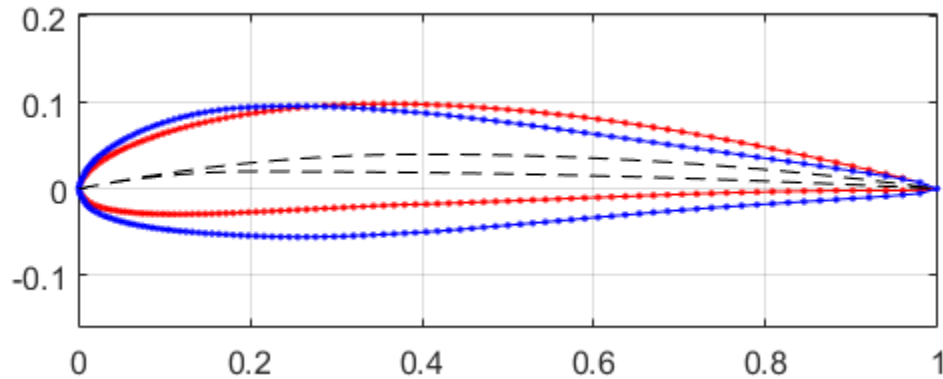


Figure 5: MATLAB plot of custom foil (blue) and NACA4412 foil (red)

Figure 6 below is the lift coefficient vs angle of attack across the range of 2 degrees to 8 degrees of both foils. My airfoil performs much worse at all angles, but performs especially worse from 2 to 3.5 degrees. This is most likely due to the thickness of the airfoil. Because the angle of attack is low, a thicker airfoil will perform worse. Also, the NACA4412 has a higher camber which most likely causes it to generate more lift. The average lift coefficient for the custom foil was 0.75863 while the average lift coefficient for the NACA4412 foil was 0.96468.

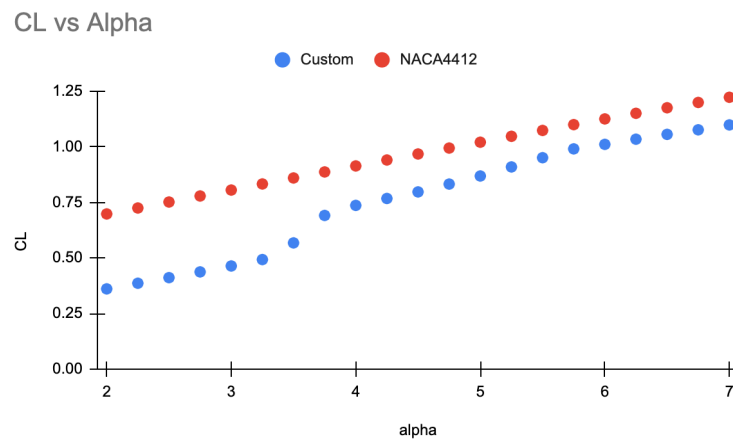


Figure 6: Lift coefficient vs Angle of Attack for a custom foil and a NACA 4412 foil.